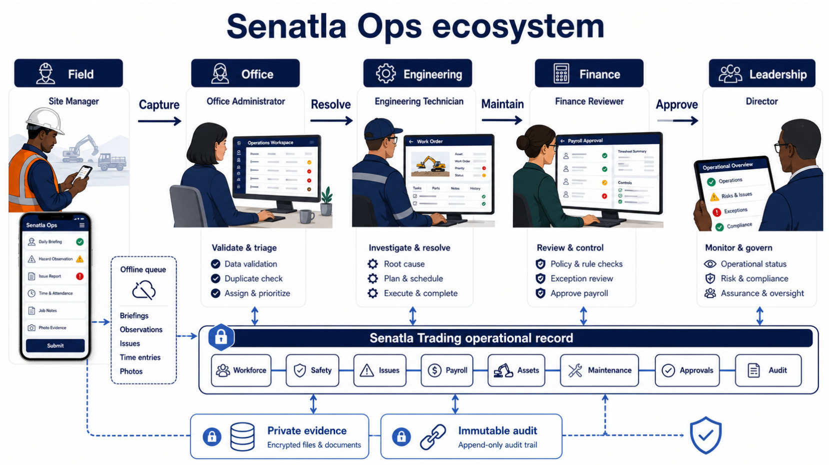
Senatla Ops

Client Ecosystem Review Pack

Ecosystem, role experiences, asset lifecycle, requirements, delivery gates and client decisions prepared by Sankofa Digital Studio.



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Prepared by Sankofa Digital Studio for Senatla Trading

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Client: Senatla Trading **Prepared by:** Sankofa Digital Studio **Version:** 1.0 **Date:** 23 June 2026 **Classification:** Confidential - client planning document

1. Executive direction

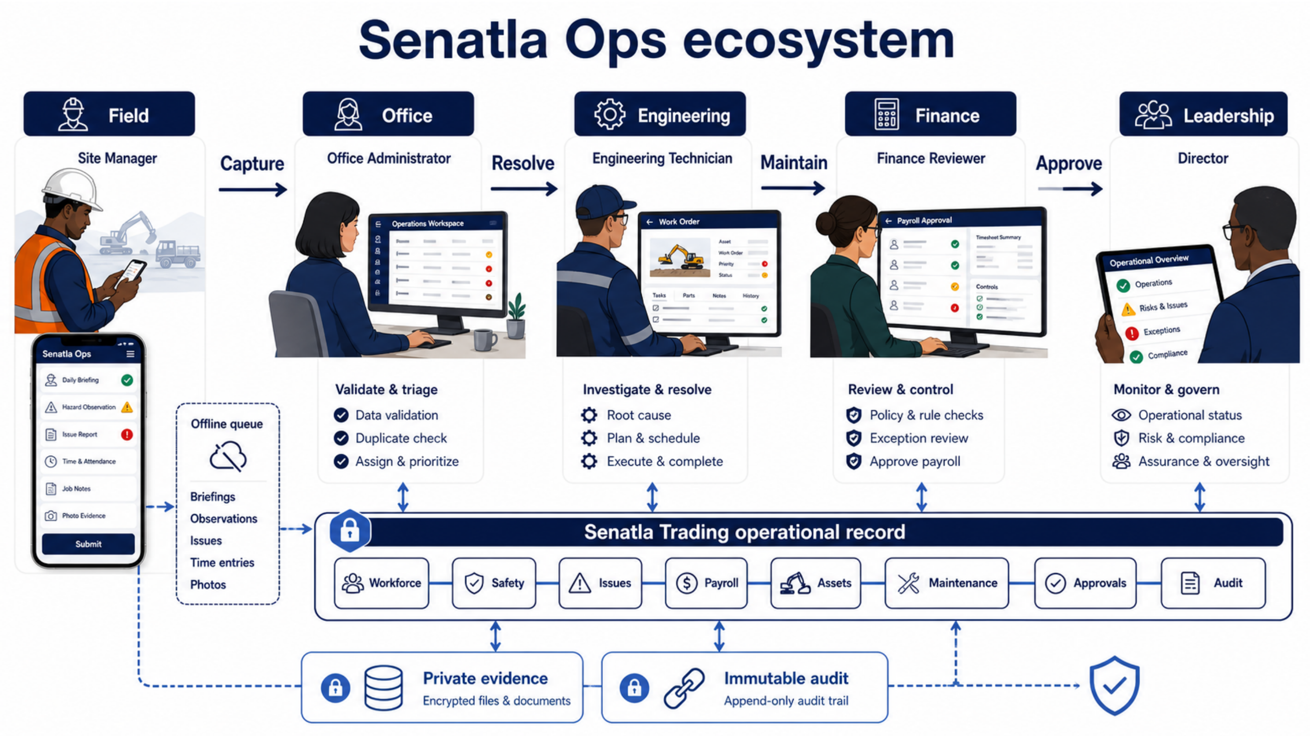
Senatla Ops should become the internal operating heartbeat for Senatla Trading. It should answer five questions without reconstructing information across spreadsheets and messages:

1. Are people present, safe and synchronized? 2. Which operational exceptions require action now? 3. Where is each asset, who is responsible for it and is it safe to use? 4. Which financial and approval controls are unresolved? 5. Can every decision be traced to an attributable source record?

The public Senatla Trading website remains frozen and outside this programme. The first controlled release has one legal owner, Senatla Trading. Engineering is a functional label until a later organizational separation decision.

2. The ecosystem at a glance

Senatla Ops is one connected operating system, not a collection of dashboards. Field teams capture events where work happens; office staff validate and resolve exceptions; engineering restores asset readiness; Finance applies sensitive controls; leadership oversees the operation through traceable evidence. All roles work against one Senatla Trading operational record.



Senatla Ops ecosystem across field, office, engineering, finance and leadership

The diagram separates responsibility without separating the source of truth. The offline queue protects field continuity, private evidence storage protects documents and images, and the immutable audit preserves who did what and when.

3. Product principles

- Evidence before appearance: every score links to source records.
- Exceptions before dashboards: urgent work appears ahead of decorative metrics.
- Offline is a normal state: queued work is visible and recoverable.
- One accountable owner: custody, approvals and actions identify a person and time.

- Rules before AI: readiness and escalation remain explainable.
- Mobile and desktop are sibling experiences, not a reduced afterthought.
- Personal information is minimized, masked and accessed only for an approved purpose.

4. Stakeholders and roles

Role	Primary objective	Main experience	Controlled actions
Site manager	Complete the site day safely and accurately	Mobile-first field workflow	Attendance, safety talk, evidence, pre-start checks and daily sync
Office administrator	Resolve exceptions and maintain official records	Desktop operational workspace	Workforce, sites, issues, payroll, assets, maintenance and requests
Director	Understand operational position and approve assigned decisions	Evidence-first executive view	Read, drill down and maker-checker review
Engineering operations	Keep heavy machinery available and safe	Asset and maintenance workspace	Custody, meters, inspections, work orders and return-to-service requests
Finance reviewer	Reconcile payroll and sensitive exports	Controlled finance workflow	Period review and assigned approvals
Technical lead	Protect delivery and recoverability	GitHub, CI, monitoring and runbooks	Release, rollback and incident coordination

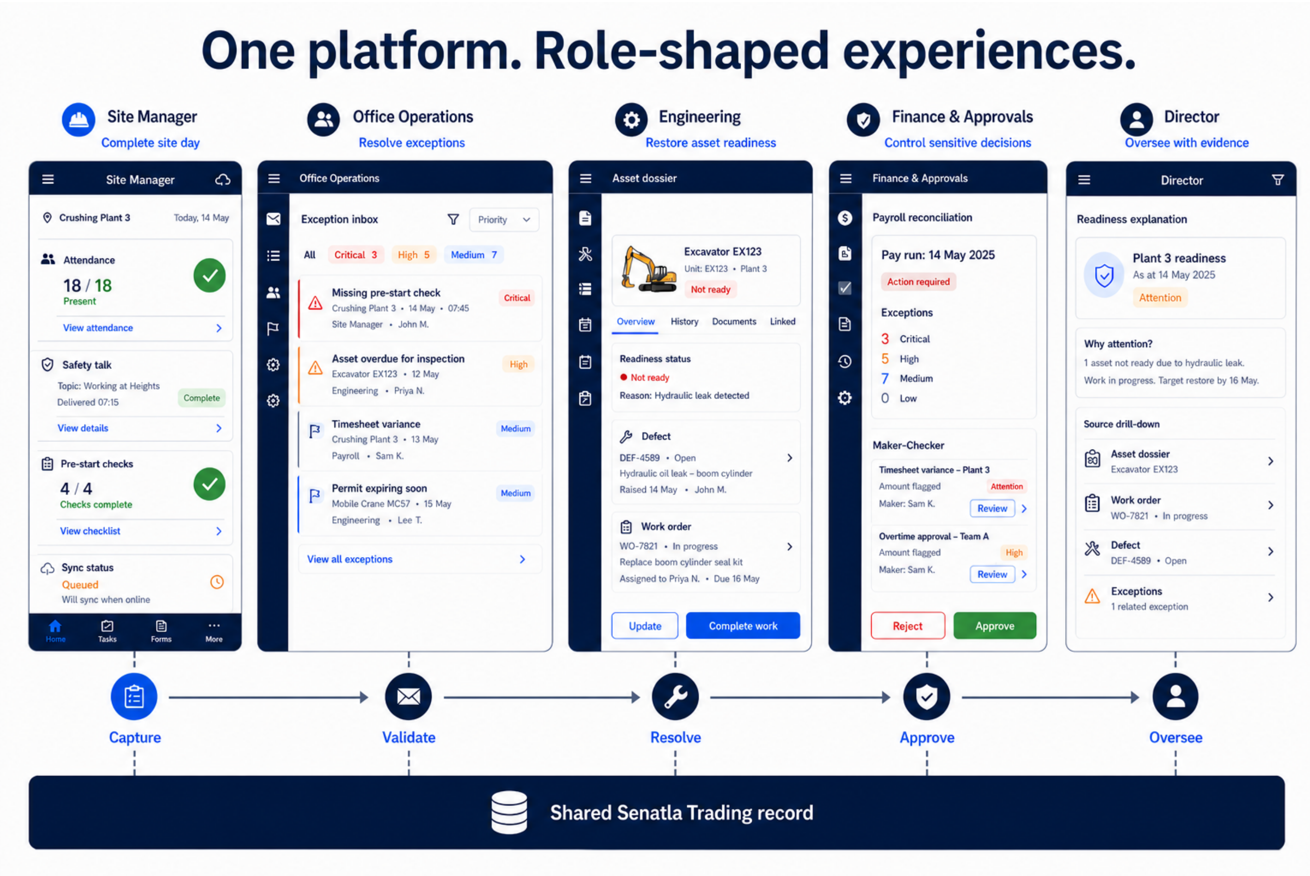
5. Functional requirements

Priority	Capability	Requirement	Acceptance evidence
Must	Authentication and access	Expired, inactive and wrong-role sessions cannot access protected workspaces	Route and RLS negative tests
Must	Daily field operations	Attendance, absence reason, safety talk and signature form one attributable daily submission	Online and offline scenario
Must	Exception control	Missing evidence, late sync, critical issue and unsafe asset create actionable exceptions	Exception-to-resolution trace
Must	Workforce and sites	Official records support validation, archive history and bulk assignment	CRUD, archive and audit tests
Must	Payroll	Calculations reconcile; locks and completed approvals are immutable	Maker-checker and reconciliation tests

Priority	Capability	Requirement	Acceptance evidence
Must	Asset identity	At least one serial, VIN or plate is required and every populated value is unique	Duplicate and normalization tests
Must	Asset lifecycle	Custody, location, compliance, meters, retirement and evidence preserve history	Transfer and retirement trace
Must	Maintenance	Failed pre-start checks create defects/work orders; critical work blocks return to service	Inspection-to-approval trace
Must	Offline delivery	Idempotency prevents duplicates and conflicts remain visible	Retry and duplicate-submission tests
Must	Audit	Every mutation creates actor, time, organization, action and entity evidence	Mutation-to-audit contract
Should	Executive evidence	Readiness, cost, downtime and compliance reconcile to persisted records	Metric reconciliation sheet
Should	QR lookup	QR opens the asset dossier but does not replace authoritative identifiers	Scan and manual fallback test
Could	Repair-versus-replace	Decision card combines downtime, repair cost, meter and replacement estimate	Explainable calculation review
Later	Spares and procurement	Parts consumption and stock support work orders	Separate approved phase
Later	Telematics and AI	External meters and prediction follow proven data quality	Separate security and ROI case

6. Role-shaped application experiences

The same platform presents a different first task to each role. The Site Manager completes the site day; Office Operations resolves exceptions; Engineering restores asset readiness; Finance controls sensitive decisions; the Director oversees with evidence. Handoffs are visible and remain linked to the shared record.



One platform with role-shaped application experiences

These concepts establish information architecture and responsibility, not final visual styling. Detailed interaction and accessibility design will be validated with users at each release gate.

6.1 Required screen family

Screen	Primary reading path	Key action
Operational command	Readiness explanation -> exceptions -> site rows -> evidence	Open highest-risk exception
Site day	Connectivity -> attendance -> safety -> inspections -> exceptions	Review and submit sync
Workforce	Search -> site/group -> employee status -> history	Update or archive official record
Issues	Severity/SLA -> owner -> evidence -> resolution history	Assign, escalate or resolve
Payroll	Period -> reconciliation -> adjustments -> approvals -> export history	Lock or request controlled export
Asset register	Identifier lookup -> dossier -> custody/compliance -> maintenance	Record lifecycle action
Maintenance	Due/critical queue -> work order -> parts/cost -> return approval	Complete safe return workflow

Screen	Primary reading path	Key action
Approvals	Request evidence -> separation check -> decision	Approve or reject with reason
Director view	Period/site -> actual values -> exceptions -> source drill-down	Review evidence, not edit records
Audit	Actor/time/action/entity -> filter -> retained evidence	Export authorized audit evidence

7. Data handling and transfer

7.1 Operational data flow

Operational role and data transfer

The client never receives a service-role credential. Supabase validates the session and RLS before data reaches Postgres or Storage. Local queues retain failed items until the server confirms the idempotency key. Readiness metrics are derived views over persisted records and never become a second source of truth.

7.2 Role transfer sequence

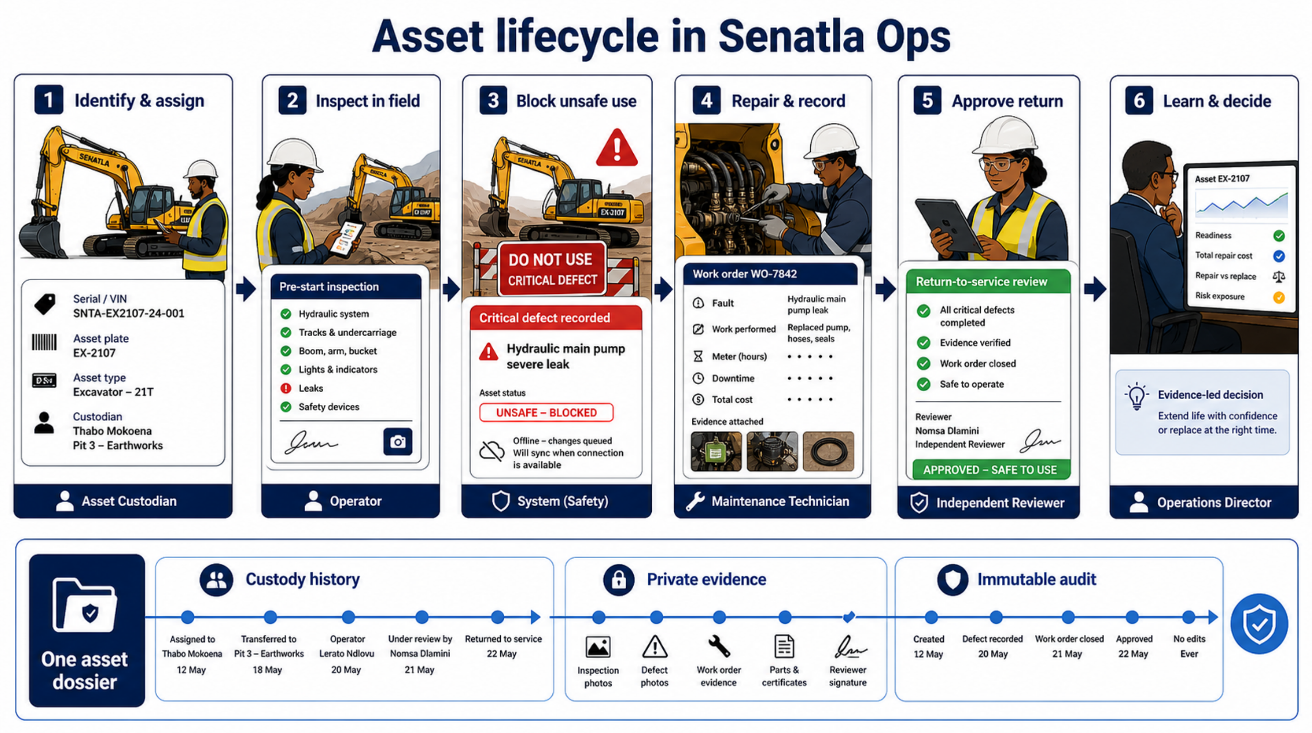
7.3 Data classification

Class	Examples	Handling guidance
Operational	Sites, issues, asset states and work orders	Authenticated access, RLS and audit
Personal	Employee identity, attendance and evidence	Purpose limitation, masking and restricted export
Financial	Rates, adjustments and payroll exports	Office/Finance access, maker-checker and immutable periods
Safety evidence	Photos, inspections and signatures	Private storage, retention policy and controlled download
Security	Sessions, audit events and incident records	Restricted access, append-only evidence and monitoring

Class	Examples	Handling guidance
Configuration	Public runtime URL and publishable key	No secrets; environment-specific generation
Secret	Service-role key and deployment credentials	Server/CI secret store only; never browser, Git or documents

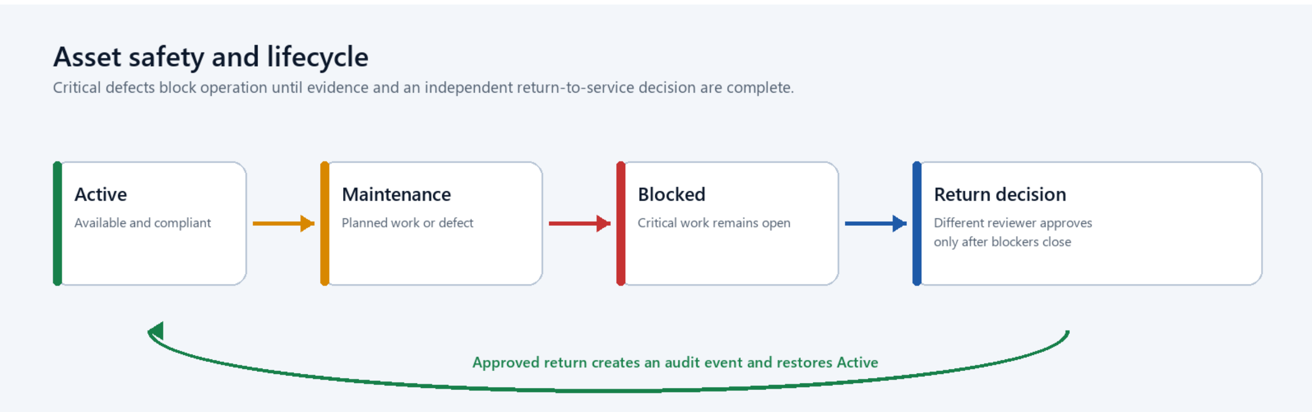
8. Asset and maintenance story

The asset workflow is one continuous story. The same excavator dossier follows identification, field inspection, safety blocking, engineering repair, independent return approval and leadership review. Custody, evidence and audit do not reset when responsibility changes.



Heavy machinery asset lifecycle across the Senatla Ops ecosystem

8.1 Controlled lifecycle states



Asset safety and lifecycle flow

Custody and meter readings are append-only. Retirement does not delete the dossier. QR codes locate the dossier; serial, VIN and plate remain the legal/operational identity controls.

9. Delivery plan and client decisions

Gate	Client-visible outcome	Sign-off
0 - Baseline	Existing work is stabilized and evidence gaps are listed	Technical lead
A - Governed	GitHub workflow, CI, environments and runbooks are active	Engineering
B - Secure data	Single-owner migration and role-negative access tests pass	Engineering/Operations
C - Core operations	Site day, workforce, issues and payroll controls pass	Operations/Finance
D - Asset pilot	Selected assets import without duplicates and lifecycle history is traceable	Engineering operations
E - Maintenance	Critical inspection through approved return to service passes	Engineering operations
F - Production candidate	Monitoring, restore, UAT and documentation are signed	Engineering/Operations/Finance

Client decisions required before Gate D are the pilot sites, pilot asset list, asset class vocabulary, custodians, evidence-retention periods and designated reviewers. These decisions should be captured as controlled configuration rather than hard-coded UI text.

10. Brainstormed enhancements

Near-term, high value

- Operational heartbeat inbox that ranks exceptions by safety, payroll and downtime impact.
- Explainable asset readiness score showing compliance, open critical work, meter due and custody acknowledgement.
- Pre-start checklist templates by asset class, with conditional evidence and automatic defect creation.
- A shift handover view that records unresolved risks, assets down and next responsible person.
- Repair-versus-replace decision card using actual downtime, work-order cost and meter history.
- Data-quality dashboard for unidentified assets, duplicate people, missing owners and stale sites.
- Low-bandwidth evidence mode that compresses images, queues uploads and shows storage status.

Medium-term, after pilot evidence

- Parts consumed per work order and minimum-stock alerts without building a full procurement ERP.
- Contractor and temporary-custody workflow with expiry and return acknowledgement.
- Warranty recovery tracker connecting work orders, warranty evidence and recovered value.
- Site-to-site asset reservation with conflict detection and transport handover.
- Scheduled compliance packs for roadworthy, insurance, certification and licence evidence.
- Notification preferences by severity and role, with escalation only after an unresolved threshold.

Challenge before investment

- Telematics should proceed only when supported equipment coverage and integration cost are known.
- Predictive maintenance requires enough trustworthy meter, failure and cost history to outperform rules.
- AI summaries must cite source records, avoid personnel scoring and never make approval decisions.
- A full spares warehouse module should be bought or integrated if an existing product meets requirements better.
- Multi-company tenancy should wait until legal ownership and cross-company data-sharing rules are approved.

11. Acceptance and guidance

A release is not accepted because it builds. Acceptance requires role-negative security evidence, a clean migration reset, desktop/mobile workflow proof, metric reconciliation and recovery rehearsal. Demo or illustrative data must be labelled. Unverified checks remain open risks.

Sankofa should maintain the source Markdown, responsive HTML, Word document and PDF as one versioned package. The engineering handbook governs implementation; this client blueprint governs expectations, decisions and acceptance.